Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

CampusXplore: A QR-Based Campus Navigation and Building Information System

Team Name: *Team #*	*Runtime Renegades* *T039*
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

CampusXplore is a C++ project designed to help students , faculty and visitors find the route between places on campus . CampusXplore uses a weigted graph to display campus location s and the distances between them and it shows all the places in a single map . In this project we use Dijkstra's algorithm to find the path , It also indicate stairs and elevatrs for moving between floors . It also lets you avoid stairs if you wish, which is helpful, for those who need it . The main aim of CampusXplore is to make finding places around the campus easier and more convenient for everyone .

Project Approach and Architecture

CampusXplore represents campus locations and distances between them using a weighted graph.
• Location class: Contains information about each location.
• Graph class: Manages all the places and the links between them.
• Weighted edges : stores the distance between places .
• Location search: it takes a source (starting point) location and destination location from the user.
• Dijkstra's algorithm: Finds a path between two points.
• Priority Queue - Processes locations in an efficient sequence. Helps dijkstra algorithm to find the shortest route.
• Route reconstruction: Project display s the shortest path, after calculation .
• Floor navigation: We connect diffrent floors by using stairs and elevators .
• Accessibility: Project offers an option to avoid stairs for users who prefer elevators or have some disability .
Project is built in C++ using STL with a console-based interface .

Tasks Completed

Task Completed	Team Member
Created the Location class to store campus details .	Garvit Aggarwal
Built the Graph class and added weighted connections .	Arnav Chamoli
Added destination search feature .	Ranveer Singh Bisht
Implemented Dijkstra's algorthm using a priority queue .	Arnav Chamoli
Added a Route Display feature .	Garvit Aggarwal
Added elevators for multifloor navigation	Raj Maletha
Distance Calculation : Verify that distances, between campus locations are calculated correctly .	Ranveer Singh Bisht

Challenges/Roadblocks

• Graph Design : We had to figure out how to connect places on campus and set the distances .
• User Input: People sometimes type location names wrong so we added checks to handle those mistakes .
• Finding the Shortest Route : We used Dijikstra's algorithm to find the path, between two places. We looked at all routes and picked the one that was the shortest , this helped us get from one place to another in the shortest way possible .
• Multi-Floor Navigation : Connecting floors was tricky so we included stairs and elevatrs in the map .
• Accessibility: We added an option that avoids stairs and finds a route if one is available .
• Edge Case Handling : We handled paths that are disconnected or places that can't be reached .
• Testing: We still need to test routes to make sure the systm works correctly in all kinds of cases .

Tasks Pending

Task Pending	Team Member (to complete the task)
Adding Locations: Add more campus locations and note their distances.	Garvit Aggarwal
Testing Routes: Ensure the system finds the correct paths .	Raj Maletha
Handling Errors: Make the program respond better to inputs .	Ranveer Singh Bisht
Floor Connectivity: Look at stair and elevatrs connections,between floors.	Arnav Chamoli
Accessibility: Make sure that avoiding stairs gives a route.	Garvit Aggarwal
Final Improvements: Fix any remaining issues . Make the program more reliable .	Raj Maletha
Documentation & Code Cleanup : Organize source files in github ,add comments in code and README file for better understanding .	Arnav Chamoli

Project Outcome/Deliverables

• Campus Graph : It is a tool designed to display locations on campus which helps people know how far from each other different spots are .
• Shortest Route : Calculates the shortest path between two locations on campus using Dijikstra's algorithm .
• Route Display : Displays the user the routes to take. It will also give them the walking distance of that route .
• MultiFloor Navigation : MultiFloor Navigation allows user s to navigate between floors with stairs and elevators .
• Accessibility : Accessibility feature was added to not use stairs when planning a route which makes navigation easier for everyone .
• Overall Outcome : CampusXplore built with C++ and data structures and algorithm helps people find their way , around the campus easier.

Progress Overview

1 . CampusXplore is being developed to help people find shortest routes around the campus .
2 . The fundamental classes for storing locations and managing connections are created in C++ .
3 . Added distances between various locations on campus to the graph .
4 . Users can select their starting point and destination to get the shortest route. Multi Floor and Accessibility features are also added .
5 . Implementation of Dijkstra's algorithm which helps the program compute and shows the total distance .
6 . The core development work is on track with more enhancements and new features to be added in the next phase .

Codebase Information

• Project Name: CampusXplore
• Programming Language: C++
• Repository Link: https://github.com/agggarvit07-lgtm/CampusXplore.git
• Branch Name: main
Important Commits :
1. Step 1 : Created the Location class to store campus detail s .
2. Step 2 : Added the Graph class. Weighted connections, between locations .
3. Step 3 : Added source . and destinaton search .
4. Step 4 : Implemented Dijkstra's algorithm to find the shortest route in the graph .
5. Step 5 : Added route reconstructon and total distance display.
6. Step 6 : Added floor navigation using stairs and elevators which can help people who are disabled .

Testing and Validation Status

Test Type	**Status (Pass/Fail)**	**Notes**
Location Creation	Completed	Campus locations are stored correctly.
Weighted Connections	Completed	Distances are assigned between connected locations.
Location Search	Completed	Users can search for valid locations .
Shortest-Path Calculation	Completed	Dijkstra's algorithm finds the shortest route .
Priority Queue	Completed	All Location s are processed by distance .
Route Display	Completed	The Route and total distance are displayed .
Multi-Floor Navigation	Implemented	Stairs and elevators connect different floors .
Accessibility Option	Implemented	Users can choose to avoid stair s .
Invalid Input Handling	Implemented	Program checks whether locations exist .
Additional Route Testing	In Progress	Many routes and accessibility cases need testing .

Deliverables Progress

Deliverable / Feature	**Status**
1 . Location Class	Completed
2 . Graph Class	Completed
3 . Weightd Connections	Completed
4 . Location Search and User Input	Completed
5 . Dijikstra's Algorithm	Completed
6 . Priority Queue	Completed
7 . Route Reconstruction and Display	Completed
8 . MultiFloor Navigation	Implemented
9 . Stairs/Elevator Connections	Implemented
10. Accessibility Option to Avoid Stairs	Implemented
11 . Expanded Campus Dataset	In Progress
12 . Additional Testing and Validation	In Progress
13 . QR Based Positioning	Pending
14 . Interactive Map Interface	Pending
15 . Dynamic Route Blocking	Pending
16 . Emergency Routing	Pending